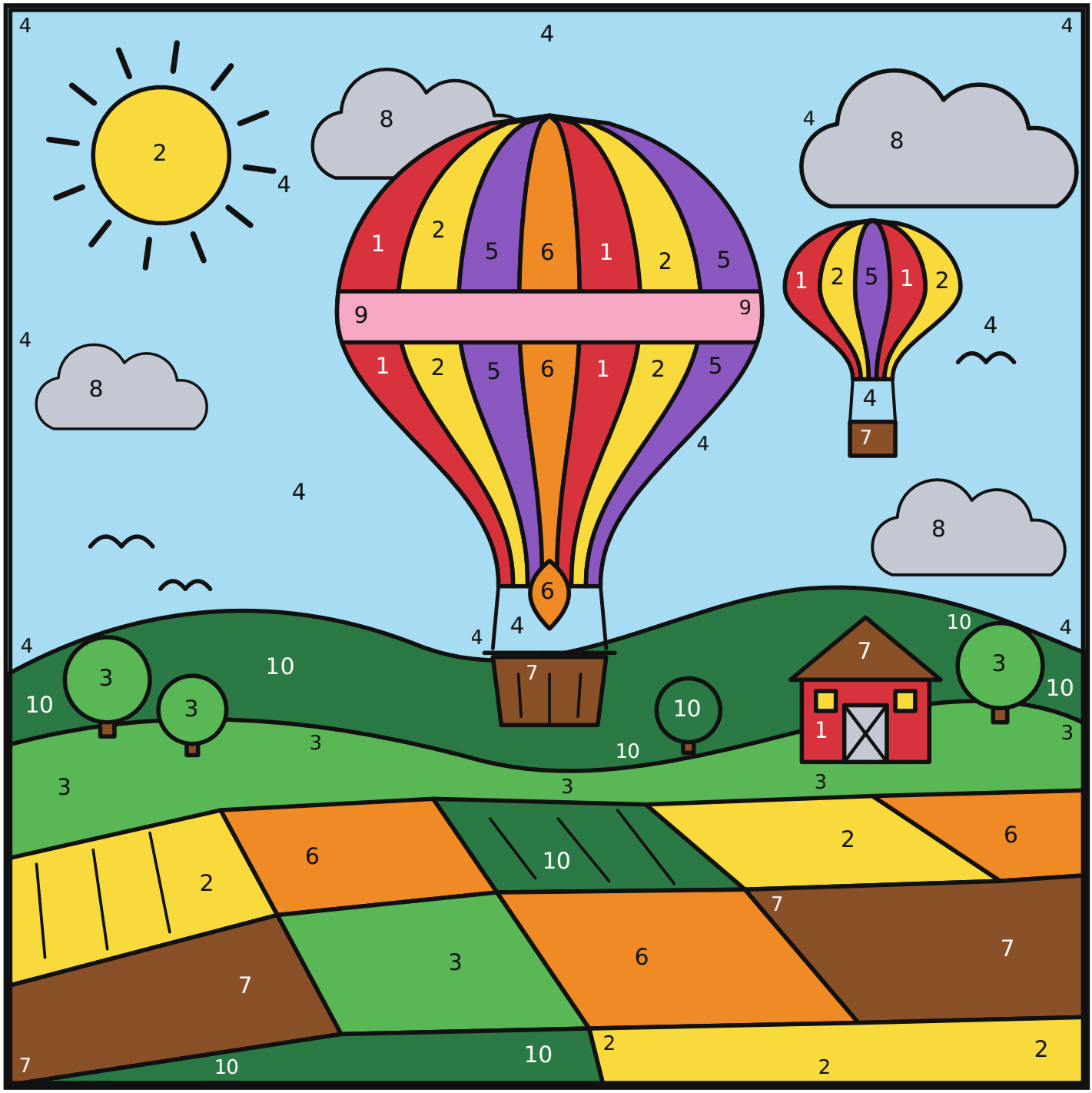
Name: **ANSWER KEY** _____ Date: _____

Linear Functions & Systems: Color By Number



Color Code:

9 Pink 4	**1** Red −7	**5** Purple 15	**10** Dark Green 24	**4** Light Blue −5
6 Orange −6	**2** Yellow 7	**8** Gray 42	**7** Brown 8	**3** Green 10

Mr. K's Classroom

Name: **ANSWER KEY** _____ Date: _____

Linear Functions & Systems: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch closely: the answers look alike!

1.
The line through $(-2, 11)$ and $(4, k)$ has slope -3. Find k.

$\frac{k-11}{4-(-2)} = -3$
$k - 11 = -3(6) = -18$
−7

1. Color: **Red**

2.
A line is perpendicular to $3x - 4y = 8$ and passes through $(6, -1)$. Find its y-intercept.

Perpendicular slope $-\frac{4}{3}$: $y + 1 = -\frac{4}{3}(x - 6)$
y-intercept: $-1 + 8 = 7$
7

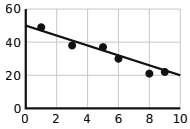
2. Color: **Yellow**

3.
A line parallel to $2x + 5y = 7$ passes through $(-5, 4)$. Written as $2x + 5y = C$, what is C?

Parallel lines keep the same x and y coefficients
$C = 2(-5) + 5(4) = -10 + 20$
10

3. Color: **Green**

4.
Notebooks left, y, after x days of a sale: $\hat{y} = -3x + 50$. On day 8, 21 were left. Find the residual (actual − predicted).



$\hat{y} = -3(8) + 50 = 26$; residual $= 21 - 26$
−5

4. Color: **Light Blue**

5.
A plumber charges a flat fee plus a fixed rate per hour. A 2 hour job costs \$95 and a 6 hour job costs \$255. What is the flat fee in dollars?

Rate $= \frac{255-95}{6-2} = 40$; $95 = 40(2) + b$, so $b = 15$
15

5. Color: **Purple**

6.
Solve by elimination: $4x + 3y = -1$ and $3x - 2y = 12$. Find the product xy.

Multiply by 2 and by 3, then add: $17x = 34$, so $x = 2$
$y = -3$, so $xy = (2)(-3)$
−6

6. Color: **Orange**

7.
For what value of k does the system $6x - 3y = 12$ and $4x - 2y = k$ have infinitely many solutions?

Divide the first equation by 3: $2x - y = 4$
Second equation must be double that: $k = 2(4)$. Any other k gives $0 =$ a nonzero number (no solution).
8

7. Color: **Brown**

8.
Scarf: 2 hours knitting, 1 finishing. Hat: 1 knitting, 2 finishing. Limits: 16 knitting, 14 finishing hours. Profit: \$5 per scarf, \$3 per hat. Find the maximum profit.

Maximize $5x + 3y$: $2x + y \le 16$, $x + 2y \le 14$
Vertices $(0, 0)$, $(8, 0)$, $(0, 7)$, $(6, 4)$: 0, 40, 21, 42
42

8. Color: **Gray**

9.
Solve: $x + y + z = 5$, $2x - y + z = 9$, $x + 2y - z = -4$. Find z.

Eq. 1 + eq. 3: $2x + 3y = 1$. Eq. 2 + eq. 3: $3x + y = 5$
$x = 2$, $y = -1$, so $z = 5 - 2 + 1$
4

9. Color: **Pink**

10.
Find the area of the shaded region (square units): $x \ge 0$, $y \ge 3$ and $3x + 4y \le 36$.



Vertices $(0, 3)$, $(0, 9)$, $(8, 3)$: base 6, height 8
24

10. Color: **Dark Green**